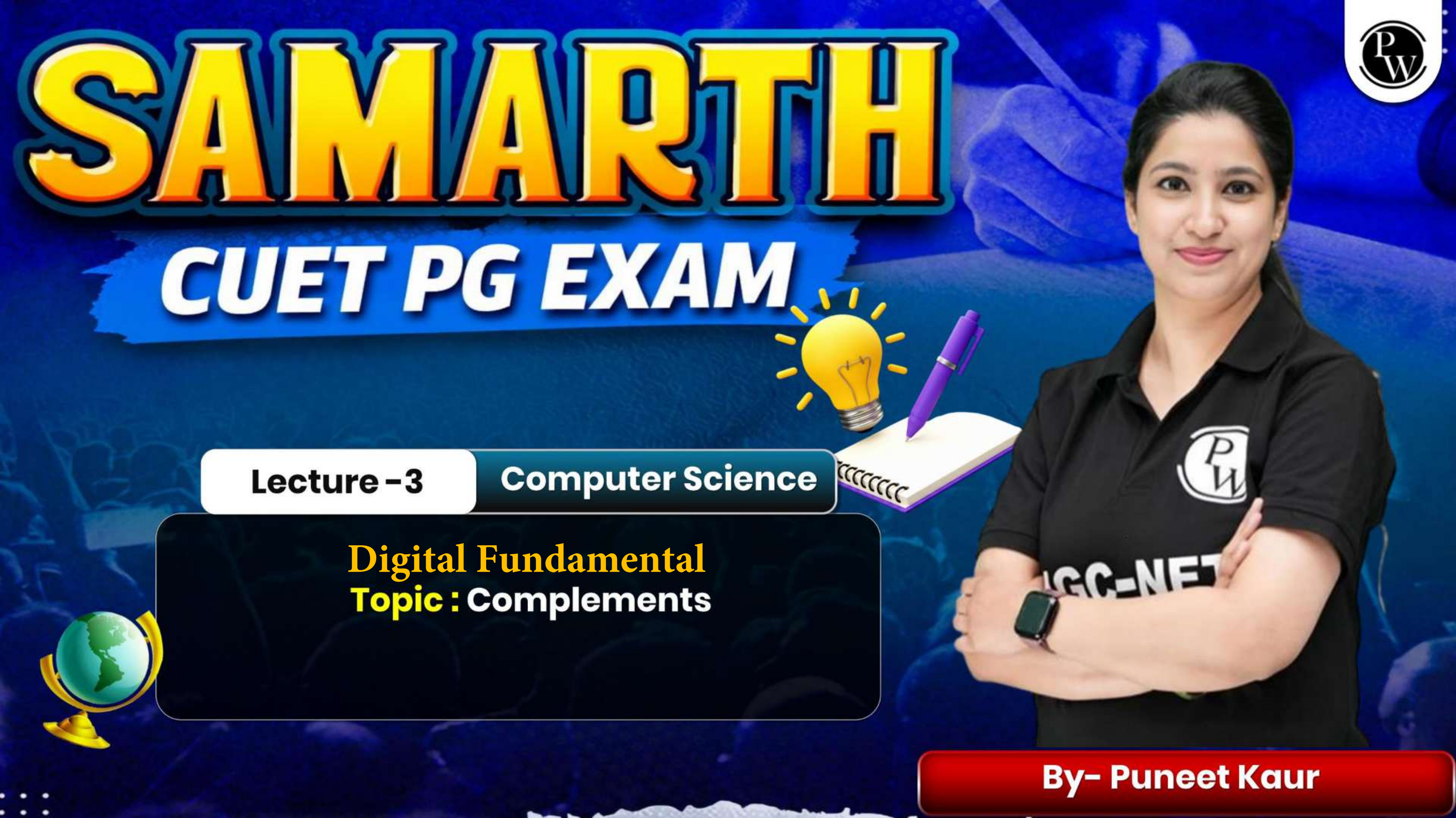
SAMARTH

CUET PG EXAM

Lecture -3 | Computer Science

Digital Fundamental

Topic : Complements



By- Puneet Kaur

Recap *of previous lecture*

1. Number System Part-2



Topics *to be covered*

1. Complements



Topic : Complements

1's Complement :

Inverting the bit (0 changed to 1 & 1 changed to 0)

Ex

1's = 1 0 1 0 1 1 → 0 1 0 1 0 0

Ex

1's = 0 1 1 1 0 0 1 → 1 0 0 0 1 1 0

2's complement :-

→ Trick ←

Ex: 10110

01010

Ex: 10111

1's 01000

+1

2's complement → 01001

Half Adder	A	B	Sum	Carry
	0	0	0	0
Add	0	1	1	0
2	1	0	1	0
bits	1	1	0	1

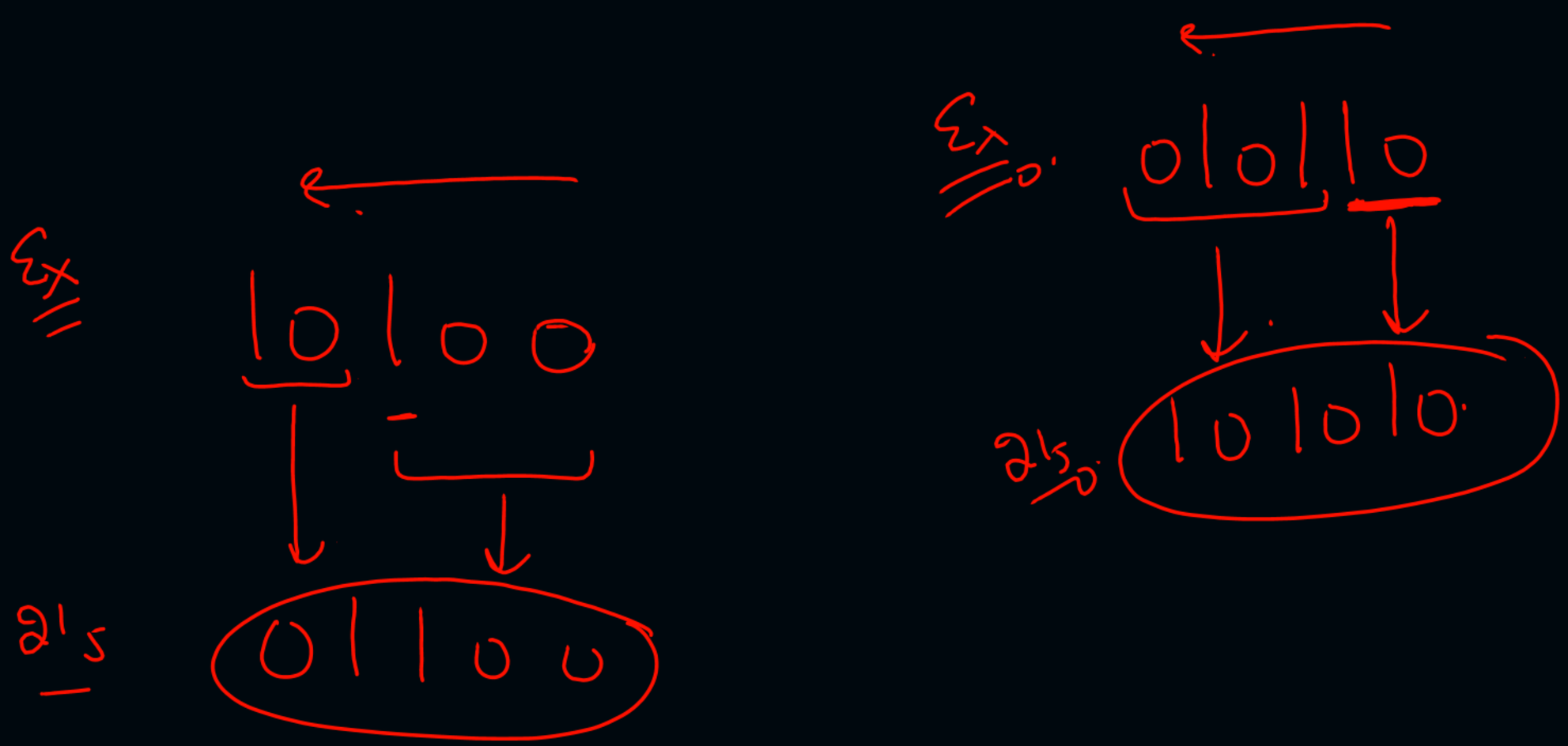
Ex: 11011

+1

11100

1+1 = 2

$= (10)_2$


Ex:1
10100
2's
01100
Ex:2.
01011O
2's
101010

7's & 8's complement

Q $\equiv (123)_8$

Max no 7

$$\begin{array}{r} 777 \\ -123 \\ \hline 654 \end{array}$$

7's complement (654)

Octal

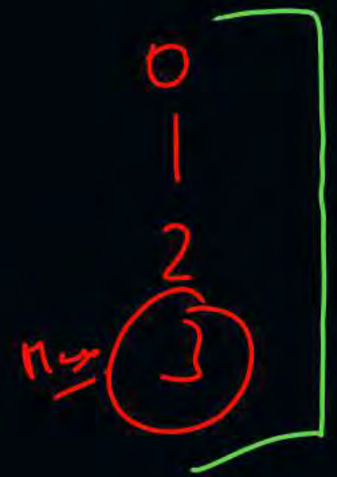
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

8's complement

$$\begin{array}{r} 654 \\ +1 \\ \hline 655 \end{array}$$

Find 3 & 4 complement.

Q $(213)_4$



$$\begin{array}{r} 3\ 3\ 3 \\ -\ 2\ 1\ 3 \\ \hline 1\ 2\ 0 \end{array}$$

3rd complement

$$\begin{array}{r} +\ 1 \\ \hline 1\ 2\ 1 \end{array}$$

4th complement

Topic : Complements

15th & 16th complement

$(123)_{16}$

15 15 15

F F F

$- 1\ 2\ 3$

15th → EDC ✓

Hex

0 – 9

A – F

11 – x

A – 10

B – 11

C – 12

D – 13

E – 14

F – 15

16th complement

EDC

+ 1

EDD

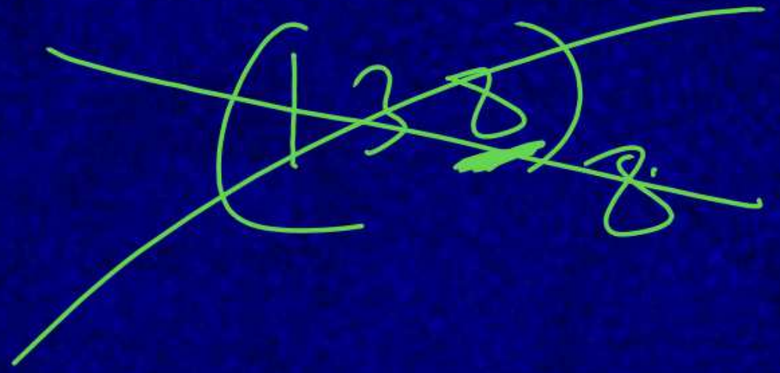
Topic : Complements

$(120)_{16}$

15 15 15
F F F.

$-1\cdot20$

E D F



Ans – (0-7)

9's & 10's complement.

Ex: 25

9's complement

$$\begin{array}{r} 99 \\ -25 \\ \hline 74 \end{array}$$

10's complement

$$\begin{array}{r} +1 \\ \hline 75 \end{array}$$

9's complement (812)

$$\begin{array}{r} 999 \\ -812 \\ \hline 187 \end{array}$$

1's
2's

7's
8's

Octal 0-7

9's
10's

1's
16's

Hex 0-9, A-F

10's

$$\begin{array}{r} 187 \\ +1 \\ \hline 188 \end{array}$$

Topic : Complements

① $(-50)_{10}$

-ve no - means find 2's complement.

2	50	
2	25	0
2	12	1
2	6	0
2	3	0
	1	1

$(50)_{10} = (110010)_2$.

-50 = 2's complement of 50

$= 110010$

$\downarrow$

001110 ✓

Q. Represent $(-20)_{10}$

2	20
2	10 — 0
2	5 — 0
2	2 — 1
	1 — 0

$(10100)_2$

$0\,|\,0\,|\,00$

$(-20)\ 1\,|\,0\,|\,00$

Q. Represent $(-5)_{10}$.

$(+5) = 0101$

sign

$(-5) = 1011$

Positive – sign 0

Negative – sign 1

$(34)_{10} = (00100010)_2$

$(-34)_{10} = (1\,0111110)_2$

Q. Represent $(-34)_{10}$ in 2's complement form.

(1) 11011100

(2) 11011111

(3) 11011110

(4) 11011101

Topic : Complements

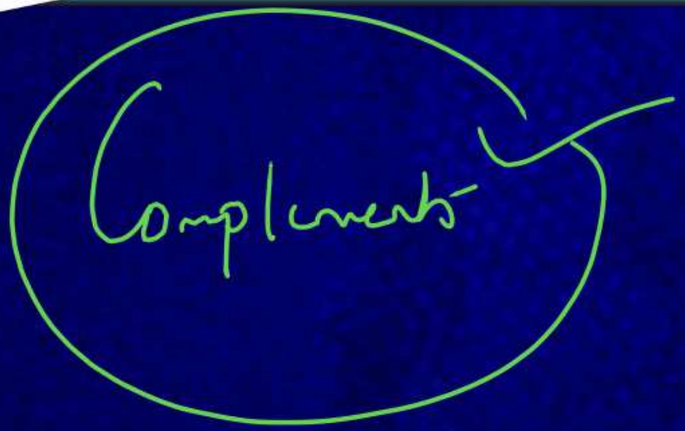
Q) Rep. $(-42)_{10}$ in 2's compl.

$42 = 0\,1\,0\,1\,0\,1\,0$

2	42	
2	21	0
2	10	1
2	5	0
2	2	1
	1	0

$-42 = 1\,0\,1\,0\,1\,0$

2 Mins Summary



Homework from SAMARTH CUET PG 2027 Module

Revision
(Summary notes)

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DPP

THANK YOU